

2022 AIME I Problems/Problem 8

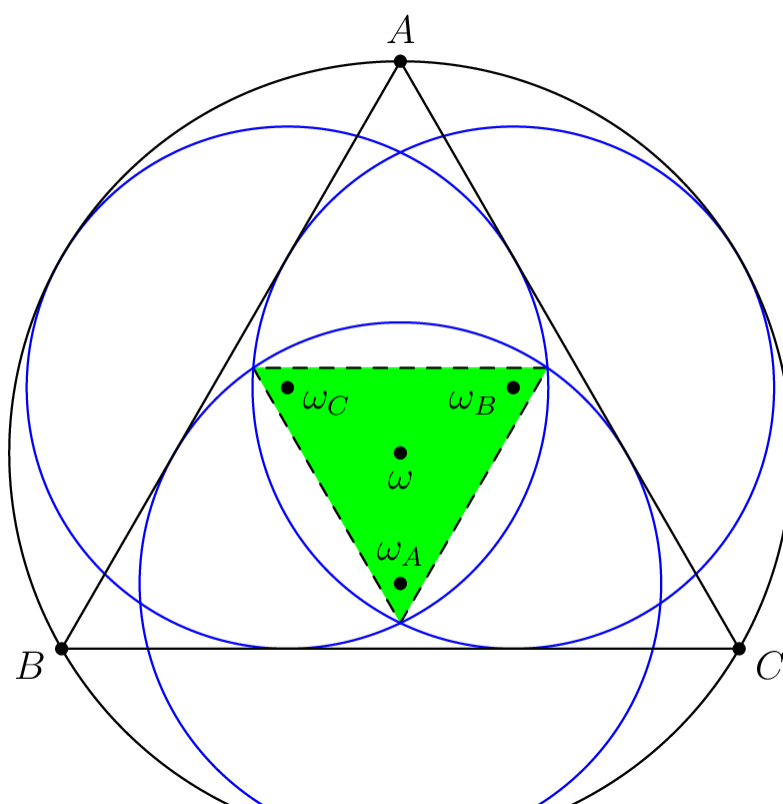
Contents

- 1 Problem
- 2 Diagram
- 3 Solution 1 (Coordinate Geometry)
- 4 Solution 2 (Euclidean Geometry)
- 5 Solution 3 (Simple Geometry)
- 6 Solution 4 (Mixtilinear Incircles)
- 7 Video Solution
- 8 Video Solution
- 9 See Also

Problem

Equilateral triangle $\triangle ABC$ is inscribed in circle ω with radius 18. Circle ω_A is tangent to sides $\overline{AB}$ and $\overline{AC}$ and is internally tangent to ω . Circles ω_B and ω_C are defined analogously. Circles ω_A , ω_B , and ω_C meet in six points—two points for each pair of circles. The three intersection points closest to the vertices of $\triangle ABC$ are the vertices of a large equilateral triangle in the interior of $\triangle ABC$, and the other three intersection points are the vertices of a smaller equilateral triangle in the interior of $\triangle ABC$. The side length of the smaller equilateral triangle can be written as $\sqrt{a} - \sqrt{b}$, where a and b are positive integers. Find $a + b$.

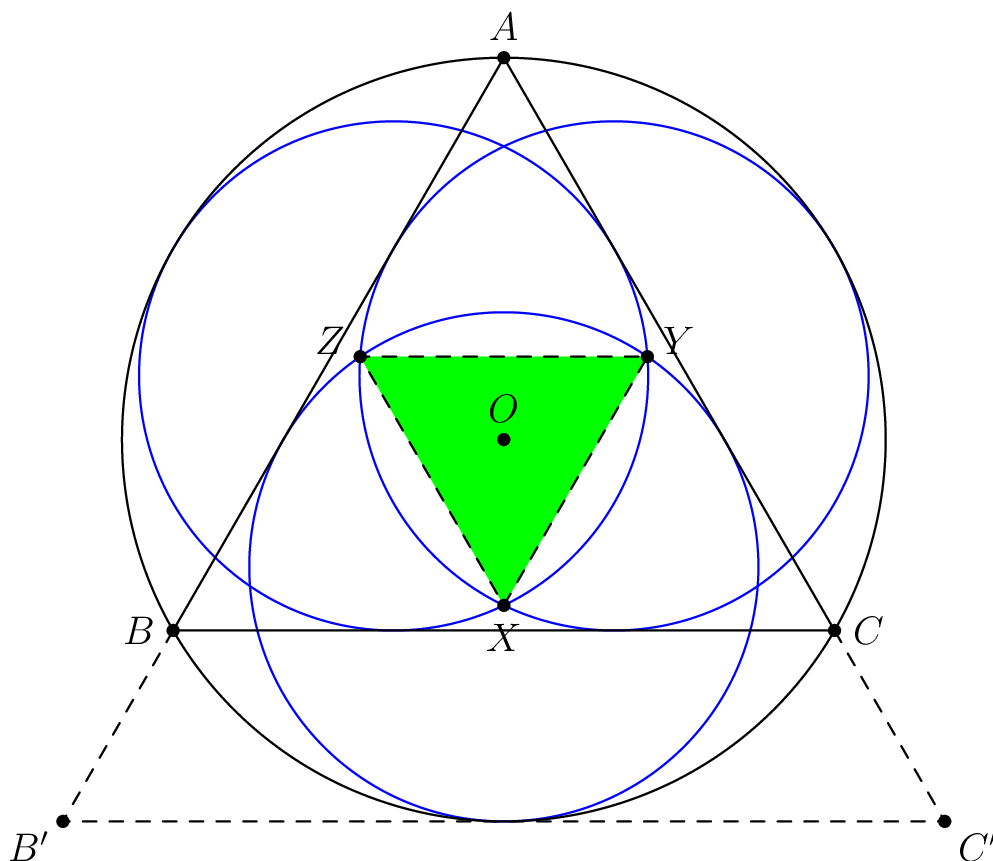
Diagram



~MRENTHUSIASM ~ihatemath123

Solution 1 (Coordinate Geometry)

We can extend AB and AC to B' and C' respectively such that circle ω_A is the incircle of $\triangle AB'C'$.



Since the diameter of the circle is the height of this triangle, the height of this triangle is 36 . We can use inradius or equilateral triangle properties to get the inradius of this triangle is 12 (The incenter is also a centroid in an equilateral triangle, and the distance from a side to the centroid is a third of the height). Therefore, the radius of each of the smaller circles is 12 .

Let $O = \omega$ be the center of the largest circle. We will set up a coordinate system with O as the origin. The center of ω_A will be at $(0, -6)$ because it is directly beneath O and is the length of the larger radius minus the smaller radius, or $18 - 12 = 6$. By rotating this point 120° around O , we get the center of ω_B . This means that the magnitude of vector $\overrightarrow{O\omega_B}$ is 6 and is at a 30° degree angle from the horizontal. Therefore, the coordinates of this point are $(3\sqrt{3}, 3)$ and by symmetry the coordinates of the center of ω_C is $(-3\sqrt{3}, 3)$.

The upper left and right circles intersect at two points, the lower of which is X . The equations of these two circles are:

$$(x + 3\sqrt{3})^2 + (y - 3)^2 = 12^2,$$

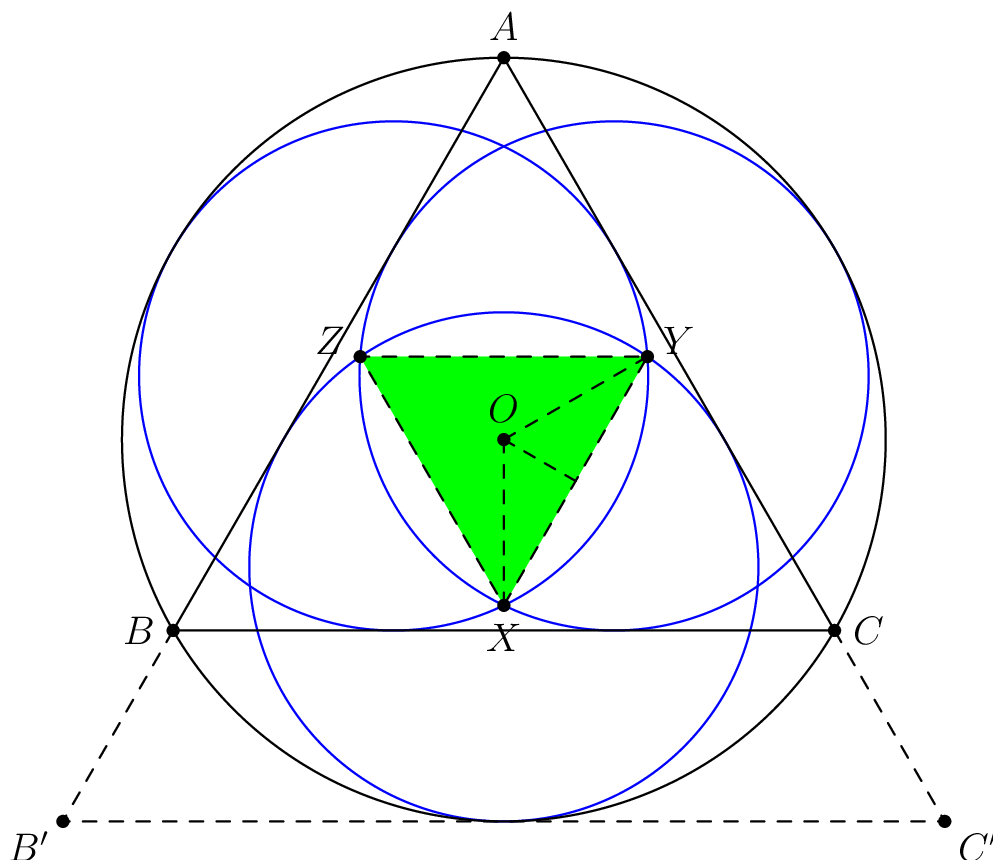
$$(x - 3\sqrt{3})^2 + (y - 3)^2 = 12^2.$$

We solve this system by subtracting to get $x = 0$. Plugging back in to the first equation, we have

$$(3\sqrt{3})^2 + (y - 3)^2 = 144 \implies (y - 3)^2 = 117 \implies y - 3 = \pm\sqrt{117} \implies y = 3 \pm \sqrt{117}$$

. Since we know X is the lower solution, we take the negative value to get $X = (0, 3 - \sqrt{117})$.

We can solve the problem two ways from here. We can find Y by rotation and use the distance formula to find the length, or we can be somewhat more clever. We notice that it is easier to find OX as they lie on the same vertical, $\angle XOY$ is 120 degrees so we can make use of $30 - 60 - 90$ triangles, and $OX = OY$ because O is the center of triangle XYZ . We can draw the diagram as such:



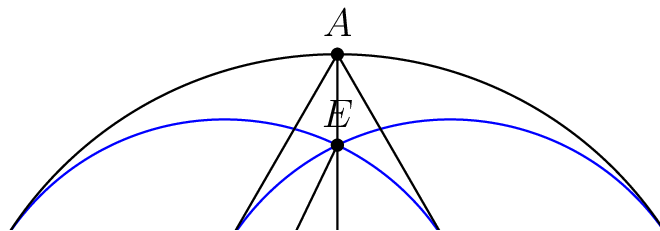
Note that $OX = OY = \sqrt{117} - 3$. It follows that

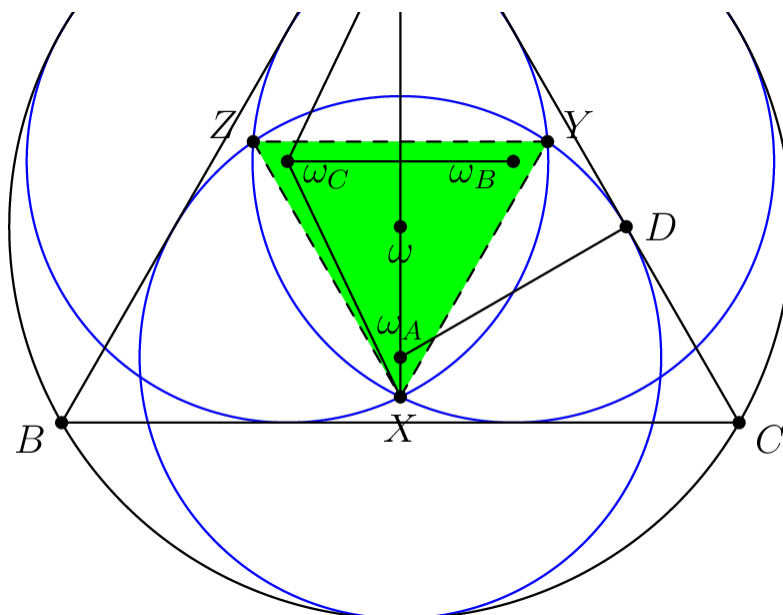
$$\begin{aligned} XY &= 2 \cdot \frac{OX \cdot \sqrt{3}}{2} \\ &= OX \cdot \sqrt{3} \\ &= (\sqrt{117} - 3) \cdot \sqrt{3} \\ &= \sqrt{351} - \sqrt{27}. \end{aligned}$$

Finally, the answer is $351 + 27 = \boxed{378}$.

~KingRavi

Solution 2 (Euclidean Geometry)





For equilateral triangle with side length l , height h , and circumradius r , there are relationships: $h = \frac{\sqrt{3}}{2}l$,
 $r = \frac{2}{3}h = \frac{\sqrt{3}}{3}l$, and $l = \sqrt{3}r$.

There is a lot of symmetry in the figure. The radius of the big circle $\odot\omega$ is $R = 18$, let the radius of the small circles $\odot\omega_A$, $\odot\omega_B$, $\odot\omega_C$ be r .

We are going to solve this problem in 3 steps:

Step 1:

We have $\triangle A\omega_A D$ is a $30 - 60 - 90$ triangle, and $A\omega_A = 2 \cdot \omega_A D$, $A\omega_A = 2R - r$ ($\odot\omega$ and $\odot\omega_A$ are tangent), and $\omega_A D = r$. So, we get $2R - r = 2r$ and $r = \frac{2}{3} \cdot R = 12$.

Since $\odot\omega$ and $\odot\omega_A$ are tangent, we get $\omega\omega_A = R - r = \frac{1}{3} \cdot R = 6$.

Note that $\triangle\omega_A\omega_B\omega_C$ is an equilateral triangle, and ω is its center, so $\omega_B\omega_C = \sqrt{3} \cdot \omega\omega_A = 6\sqrt{3}$.

Step 2:

Note that $\triangle\omega_C EX$ is an isosceles triangle, so

$$EX = 2\sqrt{(\omega_C E)^2 - \left(\frac{\omega_B\omega_C}{2}\right)^2} = 2\sqrt{r^2 - \left(\frac{\omega_B\omega_C}{2}\right)^2} = 2\sqrt{12^2 - (3\sqrt{3})^2} = 2\sqrt{117}.$$

Step 3:

In $\odot\omega_C$, Power of a Point gives $\omega X \cdot \omega E = r^2 - (\omega_C\omega)^2$ and $\omega E = EX - \omega X = 2\sqrt{117} - \omega X$.

It follows that $\omega X \cdot (2\sqrt{117} - \omega X) = 12^2 - 6^2$. We solve this quadratic equation: $\omega X = \sqrt{117} - 3$.

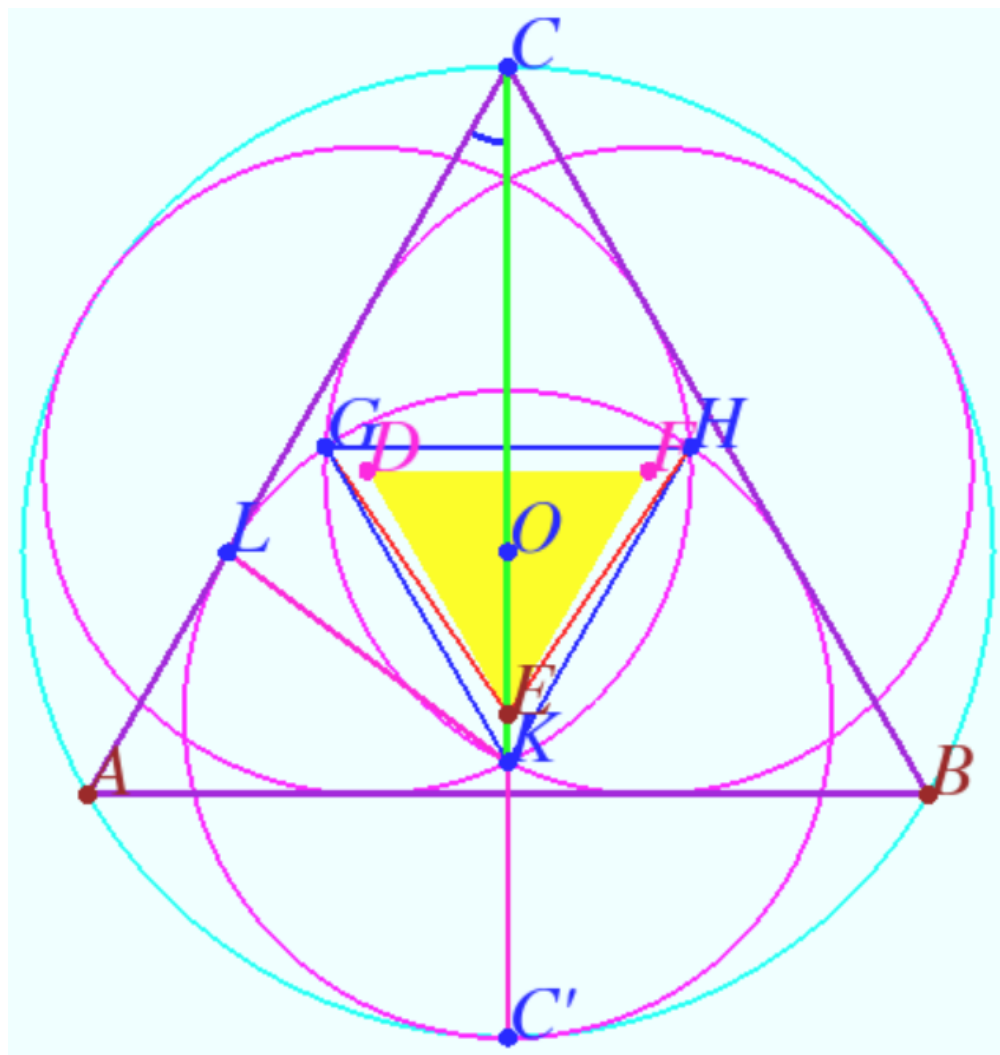
Since ωX is the circumradius of equilateral $\triangle XYZ$, we have

$$XY = \sqrt{3} \cdot \omega X = \sqrt{3} \cdot (\sqrt{117} - 3) = \sqrt{351} - \sqrt{27}.$$

Therefore, the answer is $351 + 27 = \boxed{378}$.

~isabelchen

Solution 3 (Simple Geometry)



Let O be the center, $R = 18$ be the radius, and CC' be the diameter of ω . Let r be the radius, E, D, F are the centers of $\omega_A, \omega_B, \omega_C$. Let KGH be the desired triangle with side x . We find r using

$$CC' = 2R = C'K + KC = r + \frac{r}{\sin 30^\circ} = 3r.$$

$$r = \frac{2R}{3} = 12.$$

$$OE = R - r = 6.$$

Triangles $\triangle DEF$ and $\triangle KGH$ are equilateral triangles with a common center O , therefore in the triangle OEH
 $OE = 6, \angle EOH = 120^\circ, OH = \frac{x}{\sqrt{3}}.$

We apply the Law of Cosines to $\triangle OEH$ and get

$$OE^2 + OH^2 + OE \cdot OH = EH^2.$$

$$6^2 + \frac{x^2}{3} + \frac{6x}{\sqrt{3}} = 12^2.$$

$$x^2 + 6x\sqrt{3} = 324$$

$$x = \sqrt{351} - \sqrt{27} \implies 351 + 27 = \boxed{378}$$

vladimir.shelomovskii@gmail.com, vvsss

Solution 4 (Mixtilinear Incircles)

Let O be the center of ω , X be the intersection of ω_B, ω_C further from A , and O_A be the center of ω_A . Define Y, Z, O_B, O_C similarly. It is well-known that the A -mixtilinear inradius R_A is $\frac{r}{\cos^2(\frac{\angle A}{2})} = \frac{9}{\cos^2(30^\circ)} = 12$, so in particular this means that $OO_B = 18 - R_B = 6 = OO_C$. Since $\angle O_BOO_C = \angle BOC = 120^\circ$, it follows by Law of Cosines on $\triangle OO_BO_C$ that $O_BO_C = 6\sqrt{3}$. Then the Pythagorean theorem gives that the altitude of O_BO_CX is $\sqrt{117}$, so $OY = OX = \text{dist}(X, YZ) - \text{dist}(O, YZ) = \sqrt{117} - 3$ and $YZ = \frac{O_BO_C \cdot OY}{OO_B} = \frac{6\sqrt{3}(\sqrt{117}-3)}{6} = \sqrt{351} - \sqrt{27}$ so the answer is $351 + 27 = \boxed{378}$.

~Kagebaka

Video Solution

https://youtu.be/q6_LsIAfFpl

~MathProblemSolvingSkills.com

Video Solution

<https://youtu.be/NTbdG4liCRY>

~AMC & AIME Training

See Also

2022 AIME I ([Problems](#) • [Answer Key](#) • [Resources](#))

Preceded by
[Problem 7](#)

Followed by
[Problem 9](#)

[1](#) • [2](#) • [3](#) • [4](#) • [5](#) • [6](#) • [7](#) • [8](#) • [9](#) • [10](#) • [11](#) • [12](#) • [13](#) • [14](#) • [15](#)

[All AIME Problems and Solutions](#)



These problems are copyrighted © by the [Mathematical Association of America](#).

Category:

- [Intermediate Geometry Problems](#)

Copyright © 2026 Art of Problem Solving